

INSTITUTE OF ACTUARIES OF INDIA
EXAMINATIONS

May 2026

CP1A - Actuarial Practice

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

Q.1) A conglomerate has sought advice from an external actuary about raising capital from different sources.

i) Briefly state what preliminary work the external actuary should do before giving advice. [2]

ii) Define the various types of advice that can be provided by an external actuary. [3]
[5 marks]

Q.2) A life insurance company writes only non-participating non-linked business with assets majorly invested in bonds. The yields on long term government bonds have been falling and gap in yields between government bonds and corporate bonds have been rising in the country. The company is looking to diversify the investment and looking to invest in property and equities.

i) Explain the reasons for above economic phenomenon pertaining to yields on bonds. [4]

ii) List the ways for indirect investment in property. [1.5]

iii) Briefly state the reasons for company choosing indirect investment in property as compared to direct investment. [2.5]

iv) As the yields are falling, the equity markets have gone up. Explain the reasons why the equity markets have gone up in the declining bond yield scenario. [2]

[10 marks]

Q.3) The country is observing a high level of trading in electricity between electricity generation plants and electricity distribution companies over privately owned electricity stock exchanges. The government wants to introduce a regulation which will include a requirement to obtain license to operate as an exchange.

i) Describe the regulatory requirements that an electricity stock exchange might need to fulfil to obtain the license. [5]

ii) Discuss the impact of climate change and environmental changes on the operation of electricity stock exchange. [5]

[10 marks]

Q.4) A state government is looking to establish another airport in a metro city to reduce the airport congestion and improve ease of commuting for travelers.

i) Describe the risks that might emerge from the above proposal for various stakeholders. [10]

ii) Write down the contents that might be included in risk portfolio or risk register. [5]
[15 marks]

Q.5) A life insurance company, writing all lines of business, is developing a new non-participating endowment product. The actuary wants to determine the expense loadings to be used for this product. The expense analysis performed by the company consists of following costs for all lines of business:

- Employee remuneration cost

- Lease cost
- IT cost
- Medical underwriting cost

Employee remuneration cost pertains to sales, new business data entry, underwriting, finance, actuary and other support function departments.

Describe how the expense loadings will be derived by the actuary based on above expense analysis for the new non-participating endowment product.

[8 marks]

Q.6) A defined benefit pension scheme of a company has been funded with regular contributions and the benefits are linked to final salary. As per the pension schemes, the pension payments will increase by the domestic equity returns with a guarantee of no reduction in pension payments. The investment strategy of the fund has been developed by the scheme trustees and it is reviewed on annual basis.

- i) State the investment controls that might be included by the scheme trustees. **[4]**
- ii) Describe the suitable assets matching the liabilities that might be part of the investment strategy. **[6]**
- iii) Explain the reason for regular review of investment strategy by trustees. **[5]**
- iv) The company is looking at various sources to fund the expansion strategy. One of the proposals is to utilise the funds from pension scheme and manage the fund on pay-as-you-go basis. Discuss this proposal. **[5]**

[20 marks]

Q.7) Answer the following questions:

- i) State the special terms that might be offered by a life insurance company to applicants with substandard health. **[2]**
- ii) A large reinsurance company has provided catastrophe cover to various general insurance companies. The reinsurance company is exploring to use catastrophe bonds to transfer some of this risk. Describe the mechanism of catastrophe bonds in this case. **[5]**
- iii) A life insurance company writes only annuities contract. The company is concerned about longevity risk and wants to manage this risk with the help of longevity swaps. Discuss how the longevity swaps can be used by the company to manage the risk. **[3]**
- iv) State the reasons for availing alternative risk transfer as compared to traditional reinsurance. **[2]**

[12 marks]

Q.8) The government of a country has decided to increase the capital gains tax on equity investments of individual investors. Describe the changes that might occur in the investment portfolio of individual investors. **[5 marks]**

Q.9) A defined pension scheme wants to analyse the growth in salary of members as compared to previous year. Describe how this growth experience can be analysed by the scheme.

[5 marks]

Q.10) A general insurance company writing cyber insurance has been witnessing a fall in surplus over the past 3 years.

i) Describe the reason for this fall in surplus. **[7]**

ii) State the reasons for undertaking the above analysis of surplus by the company. **[3]**
[10 marks]
